Haonan Peng

Ph.D. in Electrical and Computer Engineering | Rochester Institute of Technology | Rochester, NY 14623, USA

Personal Website | Google Scholar | hxpmet@rit.edu | +1 206-303-0607

RESEARCH INTERESTS

My research interests lie in the integration of robotic manipulation, control, perception, motion planning, and artificial intelligence, with a primary focus on surgical robots, addressing inaccuracies caused by cable-driven mechanisms, implementing distal force estimation, and developing efficient data-driven models for better closed-loop control and automation of surgical robots.

EDUCATION

Ph.D. in Electrical and Computer Engineering Adviser: Blake Hannaford
University of Washington (UW), Seattle, WA, BioRobotics Lab 09/2020-06/2026
Dissertation: AI-enhanced Closed Loop for Cable-driven Surgical Robots - Calibration, Tracking, and Haptics

M.S. in Mechanical Engineering Adviser: Blake Hannaford
University of Washington (UW), Seattle, WA, BioRobotics Lab 09/2017-08/2019

B.S. in Mechanical Engineering
Central South University (CSU), Changsha, China 09/2012–07/2016
Thesis: Hardware Design for a Measurement Device of an Ultra-long Flexible Shaft

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Manufacturing and Mechanical Engineering Technology
Rochester Institute of Technology (RIT), Rochester, NY 08/2026-Present

AWARDS, FUNDINGS, PROFESSIONAL SERVICES

- UW ECE Yang Award for Outstanding Doctoral Student (2026).
- CoMotion Innovation Gap Fund (2022). Description: the fund supports proof-of-concept and go-to-market strategy activities to advance promising innovations to the next stage of commercial development and investment.
- Fenwick & West Third Place Prize, Hollomon Health Innovation Challenge (2021). Description: this competition encourages innovative solutions for big problems the world faces today related to health.
- Runner-up for Best Project Award, Hamlyn Winter School on Surgical Imaging and Vision (2020). Description: an award as part of the final project at Hamlyn Winter School.
- Reviewer for IEEE RAL, TASE, ICRA, ISMR, EMBC, Scientific Reports, Measurement, Journal of Computational Design and Engineering, Journal of Supercomputing.
- Serve for MS Admission Triage Committee (member) and Robotics Curriculum Committee (assistant) at UW ECE

ACADEMIC EXPERIENCE

CRTK Python Controller for RAVEN-II Surgical Robot GitHub page **09/2021–Present**
- Created a general-purpose controller software for RAVEN-II based on Collaborative Robotics Toolkit (CRTK).
- Built and tested keyboard control, go-to, trajectory follow, and random sinusoid movements.
- Currently in use across three labs and four RAVEN-II robots, operating for over 150 hours without major issues.

Distal Force Estimation for RAVEN-II Surgical Robot **04/2023–Present**
- Built a parallel robotic system that can apply arbitrary controllable force to RAVEN's end-effector.
- Built motor-cable units with load cell feedback to output smooth force under passive movement.
- Developed a GRU based model that achieved ~0.5N accuracy in force estimation, without additional sensors

- Developed an easy and efficient calibration process for motor locations and forces.

Efficient Learning-based Calibration for Cable-driven Surgical Robot **09/2021–08/2023**

- Developed a calibration method for RAVEN-II's positional joints using neural networks to mitigate inaccuracies caused by cable-driven mechanisms.
- Developed regression-based calibrations that meet 1000 Hz servo control loop, with comparable accuracy.
- Calibration takes less than 21 minutes, including initialization, data collection, and training of calibration models.
- RAVEN-II has calibrated accuracy with 6 times smaller errors than the previous state-of-the-art, reducing 93% error of the positional joints. The accuracy stays within 0.771˚ and 0.246 mm during 6-hour loaded operations.
- Used removal and noisy ablation study to find joint positions, velocities, and torques important to the calibration performance. Resolved homing inconsistency so that restarts require no re-calibration.

Non-invasive Continuous Blood Pressure Estimation **09/2020–Present**

- Built and tested a prototype using ECG and PPG sensors to estimate blood pressure. Developing a learning-based approach with current accuracy of ±5 mmHg compared to arterial lines.
- Collaborating with UW Medical Center and University of Minnesota to collect a database with synchronized signals of ECG, PPG, and blood pressure from operating rooms. Currently, more than 3000 OR cases are recorded in the database.

Active Synthetic Image Generator for Surgical Instrument Segmentation **07/2020–09/2021**

- Designed a blending algorithm to generate synthetic endoscopic images based on active learning, which provides training data for a CNN-based segmentation model with less concern for domain gaps.
- The segmentation model achieved comparable performance with 50% reduced usage of real images.

Neural Network Estimator for RAVEN-II Surgical Robot **08/2018–08/2019**

- Developed and tested a neural network estimator for the cable-driven surgical robot, decreasing the RMS error of the end effector's position estimation by 83.6%.
- Established a sub-mm level visual motion tracker for RAVEN-II, using low-cost webcams, with easy calibration. Results video. This method was cited, utilized, and improved by research groups at UCB and JHU.

PUBLICATIONS

JOURNALS

J1. Reducing annotating load: Active learning with synthetic images in surgical instrument segmentation
Peng, H., Lin, S., King, D., Su, Y. H., Abuzeid, W. M., Bly, R. A., ... & Hannaford, B.
Medical Image Analysis, 2024, 97, 103246. [Link]

J2. Efficient Data-driven Joint-level Calibration of Cable-driven Surgical Robots
Peng, H., Lewis, A., Su, Y. H., Lin, S., Chiang, D., … & Hannaford, B. (2024).
Nature - npj Robotics, 2024. [Link]

J3. Ablation Study on Features in Learning-based Joints Calibration of Cable-driven Surgical Robots
Peng, H., Lewis, A., Su, Y. H., & Hannaford, B.
IEEE Transactions on Automation Science and Engineering, 2025, [Link]

J4. Automatic summarization of endoscopic skull base surgical videos through object detection and hidden Markov modeling
King, D., Adidharma, L., **Peng, H.**, Moe, K., Li, Y., Yang, Z., ... & Bly, R. A. (2023).
Computerized Medical Imaging and Graphics, 2023 [Link]

J5. Multi-Frame Feature Aggregation for Real-Time Instrument Segmentation in Endoscopic Video
Lin, S., Qin, F., **Peng, H.**, Bly, R. A., Moe, K. S., & Hannaford, B.

IEEE Robotics and Automation Letters, 6(4), 6773-6780, 2021, [Link]

J6. Local Style Preservation in Improved GAN-Driven Synthetic Image Generation for Endoscopic Tool Segmentation
Su, Y. H., Jiang, W., Chitrakar, D., Huang, K., **Peng, H.**, & Hannaford, B.
Sensors, 21(15), 5163. 2021, [Link]

CONFERENCES

C1. Real-time Data Driven Precision Estimator for RAVEN-II Surgical Robot End Effector Position
Peng, H., Yang, X., Su, Y., & Hannaford, B.
IEEE International Conference on Robotics and Automation (ICRA). 2020 [Link]

C2. A Wearable Device For Postoperative Breast Cancer Rehabilitation With Machine Learning For Motion Tracking
Zhao, Y., Adams, C. M., Davis, T., Zhao, J., O'Rourke, N., **Peng, H.**, ... & Raiti, J.
IEEE Global Humanitarian Technology Conference (GHTC) (pp. 437-440), 2022 [Link]

C3. Detection and Low-Latency Notification of Improper Backpack Posture using Deep Learning
Hung, H. Y., Millaway, G., Mustafa, S., **Peng, H.**, Geiger, A., & Raiti, J.
IEEE Global Humanitarian Technology Conference (GHTC) (pp. 433-436), 2022 [Link]

C4. Expanding the Surgical Robotics Community: An Intuitive Sim-to-Real Control Framework for Raven-II with a Budget-Friendly Gamepad Controller
Bui, M., Chalfant, N., Sun, C., Fabrega, S., **Peng, H.**, Huang, K., & Su, Y. H.
IEEE International Symposium on Medical Robotics (ISMR) (pp. 1-7), 2024 [Link]

C5. Virtual Reality-based Human-Robot Interaction for Remote Pick-and-Place Tasks
Xu, W., Huff, T., Ye, S., Sanchez, J. R., Rose, D., Tung, H., ... **Peng, H**. & Raiti, J.
ACM/IEEE International Conference on Human-Robot Interaction, 2024 [Link]

C6. A Disturbance in the Force: Force Actuation on the RAVEN II Surgical Robot with Parallel Motor-Cable Units
Peng, H., Chiang, D., Hendricks, J., Lewis, A., Feng, H., Su, Y. H., & Hannaford, B.
IEEE International Conference on Robotics and Automation (ICRA). 2025. ELSR Workshop. [Link]

C7. AMBF-Based Simulation of RAVEN-II Surgical Robot with Realistic Control and Feedback API
Li, B., **Peng, H.**, Chalfant, N., Su, Y. H., & Hannaford, B.
IEEE International Conference on Robotics and Automation (ICRA). 2025. ELSR Workshop. [Link]

C8. From Speech to Delivery: Human-Centered Multimodal Interaction for Indoor Service Robotics
An, A., Gajarla, H., ... **Peng, H**. & Raiti, J.
ACM/IEEE International Conference on Human-Robot Interaction, 2026. [Link]

PATENTS

P1. Cable-Driven Force Actuation System
Peng, H., Chiang, D., Hendricks, J., Shing, J., Feng, H., Lewis, A., Su, Y. H., Hannaford, B.
International PCT Patent Application Filed (2025), PCT/US2025/052314

P2. Prediction of Hemodynamic Parameters from Sensor Data Using Machine Learning
Hou, Y., **Peng, H.**, Jense, R., Reinhall, P., Johnson, M.
US provisional patent application (2024), UW Ref. 49629

P3. Preprocessing of Sensor Data for Input to Machine Learning Model
Hou, Y., **Peng, H.**, Jense, R., Reinhall, P., Johnson, M.
US provisional patent application (2024), UW Ref. 50155

PRESENTATIONS

- AI-enhanced Control for Surgical Robots, University of Alberta, 08/05/2025, Virtual
- Force Actuation on the RAVEN II Surgical Robot, ICRA 2025, 05/19/2025, Atlanta, GA
- Simulation of the RAVEN II Surgical Robot with Realistic API, ICRA 2025, 05/19/2025, Atlanta, GA
- AI-enhanced Control for Surgical Robots, Northern Arizona University, 04/01/2025, Flagstaff, AZ
- Introduction to RAVEN Surgical Robot, ControlX Seminar, 10/11/2024, Seattle, WA
- Learning-based Calibration for RAVEN-II Surgical Robot, ICRA 2020, 05/31/2020, Virtual

TEACHING AND MENTORSHIP EXPERIENCE

TEACHING

Instructor, ECE & GIX, UW

EE 543 - Models of Robot Manipulation 01/2023-03/2023, 01/2025-03/2025

- Taught this graduate-level course that covered homogeneous transformation, forward & inverse kinematics, differential kinematics, motion planning, and teleoperation.
- Created 300+ lecture slides for 10-week lectures associated with the textbook.
- Developed a project that guides students to build robot arms using servo motors and apply Cartesian control. Examples of students' work, shared with permission.
- **Course evaluation: 4.8** (out of 5) combined median.

TECHIN 512 - Introduction to Sensors and Circuits 09/2025-Present

- Created new course slides and syllabus, which provides better connection from the lecture to labs.
- Developed the Motor and Feedback Control Lab, which benefits students in their future robotics courses.
- Advised final projects on developing embedded powered handheld electronic games using microcontrollers, sensors, screens, and LEDs. Examples of students' work, open-sourced.
- **Midterm course evaluation: 4.7 and 4.8** (out of 5) on communication regarding course requirements and responsiveness to student questions, respectively.
- **Course evaluation: 4.5** (out of 5) combined median.

Teaching Assistant, ECE & GIX, UW

TECHIN 512 - Introduction to Sensors and Circuits 2020-2024

- Worked as TA **6 times in 5 years** and participated in course iterations, including creating labs, projects and quizzes.
- Helped students from non-engineering backgrounds, for whom TECHIN 512 were their first engineering course.
- Developed PPG Sensor Lab. Led labs including circuit analysis and sensor implementation.

TECHIN 517 – Robotics Lab 2 03/2023-06/2023, 03/2024-06/2024, 03/2025-06/2025

- A graduate project-based course, in which AI, perception, navigation, and manipulation of robots were developed.
- Prepared and led labs of the Fetch robot, including base & arm control, navigation, and motion planning, utilizing Gazebo, RViz, and MoveIt. GitHub page.
- Advised student project teams, resulting in **publication C5.**

TECHIN 514 – Hardware/Software Lab 1 01/2022-03/2022

- Created the first version of Wireless Lab, Power Management Lab, and LoRa ESP32 Lab.
- Advised student project teams, resulting in **publication C2, C3.**

TECHIN 515 – Hardware/Software Lab 2 03/2022-06/2022

- Led labs and advised student project teams. One of the teams won UW CoMotion Innovation Gap Funding.

EE 241 - Programming for Signal and Information Processing Applications 01/2024-03/2024

- Included code check interview to ensure students' learning experience and outcomes, while allowing AI tools

EE 447 - Control System Analysis I 09/2020-12/2020, 03/2021-06/2021

MENTORSHIP

- Dun-Tin Chiang, M.S., UW; 2023-2025, Cable-driven force actuation system, Publication: J2 C6, Patent: P1
  Thesis: Sensorless Contact Force Estimation on Raven II Surgical Robot Platform [Link]
- Justin Su, M.S., UW; 2025-, Advanced motion planning for the RAVEN-II surgical robot
  Thesis (in progress): Sim-to-Real Advanced Motion Planning for Raven II Surgical Robot
- Armina Moghadasi, B.S., UW; 2024-, Development of robot manipulation course project
  Award: Mary Gates Research Scholarship
- Jared Shing, M.S., UW; 2023-2024, Python controller for RAVEN-II surgical robot, Publication: C6, Patent: P1
- Jordan Hendricks, M.S., UW: 2023-2025, Controller for force actuation system, Publication: C6, Patent: P1
- Haokun Feng, M.S., UW; 2023-2024, Cable-driven force actuation system, Publication: C6, Patent: P1
- Qiana Chen, M.S., UW; 2024-, Customer discovery for non-invasive blood pressure device
- Wenfan Jiang, M.S., University of Michigan; 2023, Reinforcement learning on coverage trajectory, Publication: J2
- Matthew Klein, M.S., UW, 2023, ROS2 for RAVEN-II
- Helen Lai, B.S., UW, 2023, PCB design for RAVEN-II external joint encoders, Publication: J2
- Yifang Hou, M.S., 2022-2023, Non-invasive continuous blood pressure estimation, Patent: P2, P3
- Tintin Zhang, M.S., UW, 2022-2023, Business research for continuous blood pressure
- Boyang Li, B.S., UW, 2024-, Simulation of RAVEN-II with realistic control API, Publication: C7
- Anshul Prakash, M.S., UW; 2025-, Optimization of coverage trajectory in high dimensional space
- Harshavardhan R. Gajarla, M.S., UW; 2025-, Data cleaning on vital waveforms in operating rooms
- Athena Cheung, M.S., Arizona State University; 2025-, Simulation of RAVEN-II surgical robot
- Rutuja R. Mandapmalv, M.S., UW; 2025-, Attention model on feature importance in RAVEN-II calibration
- Zhenghao Jing, M.S., UW; 2025-, ROS2 and cloud data collection for RAVEN-II
- Beth Ding, B.S., UW; 2025-, Advanced control of cable-driven force actuation system (tentative)

START-UP

Co-founder, Pressurge Inc (initiative) **08/2024–Present**

- A start-up aims to commercialize AI-based non-invasive blood pressure monitoring devices for operating rooms.
- Collaborating with UW CoMotion, UW Medical Center, UW Department of Mechanical Engineering, and the University of Minnesota.
- Currently under negotiations with venture companies.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

President of Student Union, College of Mechanical and Electrical Engineering, CSU **09/2013–09/2015**

- Led the student union with 8 departments and 118 members.
- Organized the graduation ceremony and science & technology festival of the college.

34th CSU Volunteer Teaching in Zuojialong District, CSU **09/2014–12/2014**

- Organized and participated in this event, in which students of CSU provided volunteering tutoring for primary and middle school students in the local area.

Undergrad Counselor Intern, College of Mechanical and Electrical Engineering, CSU **09/2015–04/2016**

- Provided academic, mental, and safety counseling to undergraduate students.
- Supervised the student union and provided training on news and report writing.

SKILLS

Robotics: Robot manipulation, serial/parallel kinematics & dynamics, robot testing, robot maintenance, Robot Operating System (ROS), Gazebo, RViz, MoveIt, robot navigation, localization and mapping

Coding: Python, C++, MATLAB

Machine Learning: TensorFlow, Keras, PyTorch, scikit-learn, Pandas

Mechanical Engineering: SolidWorks, ADAMS, AutoCAD, finite element analysis
Perception: Encoder, strain gauge, load cell, infrared sensor, computer vision
Prototyping: Arduino, Raspberry Pi, ESP 32, 3D printing, manual and oven soldering